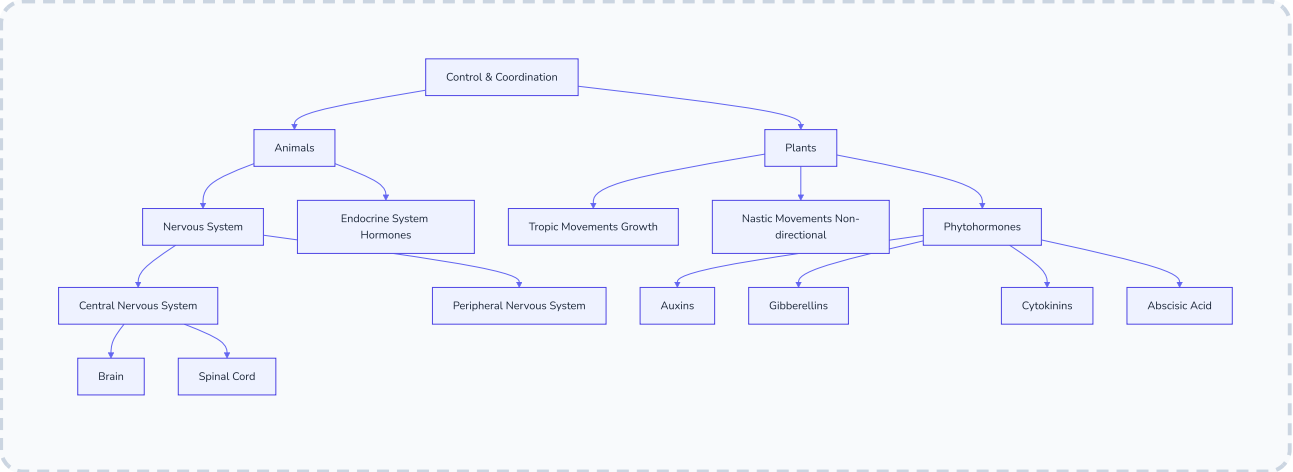
Chapter 6: Control and Coordination 🧠 ⚡

Welcome to Chapter 6 of Class 10 Science! In this chapter, we will dive deep into how our bodies and plants manage to stay in sync. How do you pull your hand away from a hot pan? How do plants know which way to grow to catch the sunlight? The answer lies in **Control and Coordination**.

Master Mind Map 🗺️



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1. Nervous System in Animals 🧬

The nervous system controls and coordinates all activities of the body by transmitting electrical signals (nerve impulses).

A. The Neuron (Nerve Cell) 🔌

The neuron is the structural and functional unit of the nervous system.

Parts of a Neuron:

1. **Dendrite**: Receives the information (electrical signal) from other neurons.
2. **Cell Body (Soma)**: Contains the nucleus and cytoplasm.
3. **Axon**: Long fiber that carries the impulse away from the cell body.
4. **Nerve Endings**: Transmits the impulse to the next neuron or target muscle.

> **QUICK TIP**
>
> Information travels as an **Electrical Impulse** within the neuron. But between two different neurons, it travels as a **Chemical Signal**.

B. Synapse

The tiny gap between two adjacent neurons is called a synapse. At the synapse, the electrical signal is converted into a chemical signal (neurotransmitter), which crosses the gap and starts a new electrical electrical impulse in the next neuron.

C. Reflex Action and Reflex Arc

A **Reflex Action** is an automatic, sudden, and involuntary response of the body to a stimulus (e.g., pulling your hand away immediately when you touch a hot pan).

Reflex Arc: The pathway taken by the nerve impulses in a reflex action. Stimulus $\rightarrow$ Receptor Organ (Skin) $\rightarrow$ Sensory Neuron $\rightarrow$ Spinal Cord (Relay Neuron) $\rightarrow$ Motor Neuron $\rightarrow$ Effector Organ (Muscle) $\rightarrow$ Response

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2. Human Brain

The brain is the highest coordinating center in the body. It is protected by a bony box called the **cranium** (skull) and surrounded by **cerebrospinal fluid** which provides shock absorption.

The Brain is divided into 3 main parts:

I. Forebrain (Thinking Part)

- **Cerebrum**: The largest part. It controls voluntary actions, memory, intelligence, hearing, smell, and sight. It is the main thinking part of the brain.

II. Midbrain

Controls involuntary actions like changes in pupil size, reflex movements of head, neck, and trunk in response to visual and auditory stimuli.

III. Hindbrain

Consists of three parts:

1. **Cerebellum**: Controls posture, balance, and precision of voluntary movements (e.g., riding a bicycle, picking up a pencil).
2. **Pons**: Regulates respiration.
3. **Medulla**: Controls involuntary actions like heartbeat, blood pressure, salivation, and vomiting.

> IMPORTANT POINTS 📌
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> The spinal cord is protected by the vertebral column (backbone) and controls reflex actions.

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3. Coordination in Plants 🌿

Unlike animals, plants do not have a nervous system or muscles. They coordinate their behavior using chemical substances (hormones) and movement in response to stimuli.

A. Tropic Movements (Directional Growth) 🌻

Tropic movements are directional growth movements in response to a directional stimulus.

- **Phototropism**: Movement towards light (e.g., Shoot bends towards light).

- **Geotropism**: Movement in response to gravity (e.g., Roots grow downwards).
- **Hydrotropism**: Movement in response to water (Roots grow towards water).
- **Chemotropism**: Movement in response to chemicals (e.g., Growth of pollen tube towards the ovule).

B. Nastic Movements (Non-directional) 🌱

Movements that do NOT depend on the direction of the stimulus. Example: Drooping of leaves of the "Touch-me-not" plant (*Mimosa pudica*) when touched. This happens rapidly due to changes in water content in plant cells (turgor pressure), not by growth.

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4. Plant Hormones (Phytohormones) 🧪

Chemical compounds that help coordinate growth, development, and responses to the environment.

Hormone	Main Function	Nature
Auxin	Promotes cell elongation, helps shoot bend towards light. Synthesized at shoot tips.	Growth Promoter
Gibberellins	Helps in the growth of the stem and seed germination.	Growth Promoter
Cytokinins	Promotes rapid cell division. Found in high concentration in fruits and seeds.	Growth Promoter
Abscisic Acid	Inhibits growth, causes wilting of leaves, and promotes seed dormancy.	Growth Inhibitor

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5. Endocrine System in Animals (Hormones) 🩸

Animal hormones are chemical messengers secreted directly into the bloodstream by **endocrine glands** (ductless glands). They act on specific target organs.

Important Human Hormones and Glands

Gland	Hormone	Function	Deficiency Disease
Thyroid	Thyroxine	Regulates metabolism of carbs, fats, and proteins.	Goitre (swollen neck) due to lack of Iodine.
Pituitary	Growth Hormone (GH)	Regulates growth and development of the body.	Dwarfism (less GH) / Gigantism (excess GH).
Pancreas	Insulin	Regulates (lowers) blood sugar level.	Diabetes.
Adrenal	Adrenaline	Prepares the body for emergency situations (Fight or Flight).	-
Testes (Male)	Testosterone	Controls male sexual development and changes during puberty.	-
Ovary (Female)	Estrogen	Controls female sexual development and changes during puberty.	-

⚠ COMMON MISTAKES ❌

> Do not confuse **Nervous Control** with **Endocrine Control**. Nervous control is extremely fast but short-lived (acts only where nerves reach). Endocrine control (hormones) is slower but long-lasting and reaches all cells via blood.

Quick Revision Notes 📌

1. **Receptors**: Specialized tips of nerve fibers that detect information (e.g., Photoreceptors for light, Gustatory for taste, Olfactory for smell).
2. **Reflex Arc**: The shortest path traveled by an impulse during a reflex action.
3. **Cerebellum**: Responsible for precision and balance. **Cerebrum**: Responsible for memory and voluntary actions.
4. **Auxin** diffuses towards the shady side of the plant, causing cells there to grow longer, making the plant bend towards the light!
5. **Adrenaline** makes the heart beat faster, increasing oxygen supply to muscles to prepare for fighting or running away.
6. **Iodine** is strictly necessary for the thyroid gland to make thyroxine hormone.

Chapter Summary 🏁

Control and coordination in animals are executed by the nervous system and endocrine system. A neuron transmits electrical impulses, while synapses convert them to chemical signals momentarily. Reflex actions protect us fast without brain interference. In plants, phytohormones manage growth directions (tropisms) and cell division. Endocrine glands release hormones into the blood to manage slower, long-term functions like metabolism, sugar regulation, and growth in human beings.